



Tip: the journey bar is clickable - jump back to any finished stage, or between your tests, anytime.



RESULTS

# Results

01 · LOGISTIC REGRESSION

predict a yes/no outcome

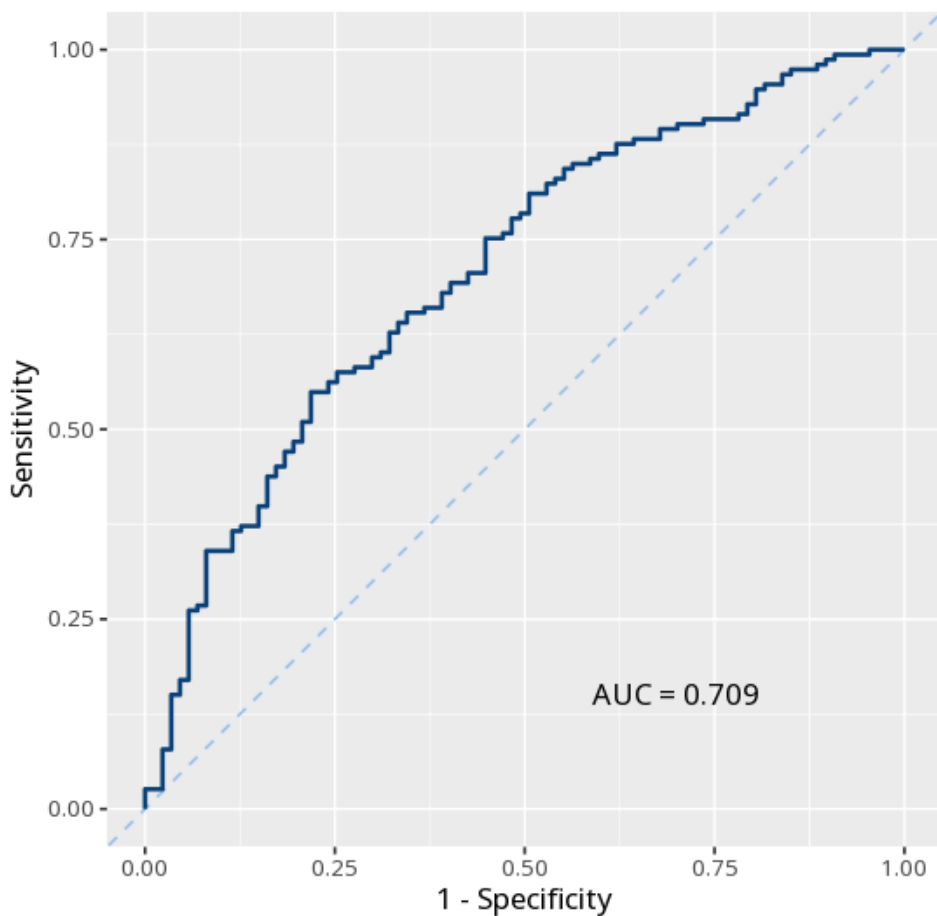
Table 1. Coefficients

|                           | Log-odds (B)     | Odds ratio (OR)        |
|---------------------------|------------------|------------------------|
| (Intercept)               | −2.69<br>(0.96)  | 0.07<br>[< 0.01, 0.43] |
| pretest_score             | 0.04<br>(0.02)   | 1.04<br>[1.01, 1.08]   |
| study_hours_per_week      | 0.19<br>(0.06)   | 1.21<br>[1.08, 1.37]   |
| teaching_method: Flipped  | 0.96<br>(0.37)   | 2.61<br>[1.27, 5.52]   |
| teaching_method: Lecture  | −0.50<br>(0.33)  | 0.60<br>[0.31, 1.16]   |
| Num.Obs.                  | 240              |                        |
| Nagelkerke R <sup>2</sup> | 0.17             |                        |
| Omnibus $\chi^2$          | 31.41 (p < .001) |                        |
| Log.Lik.                  | −141.46          |                        |
| AIC                       | 292.91           |                        |
| BIC                       | 310.31           |                        |
| Accuracy                  | 69.2%            |                        |
| Sensitivity               | 86.9%            |                        |
| Specificity               | 37.9%            |                        |
| AUC [95% CI]              | .71 [.64, .78]   |                        |

Table 2. Classification

| Predicted \ Observed | No | Yes | % correct |
|----------------------|----|-----|-----------|
| No                   | 33 | 20  | 62.3%     |
| Yes                  | 54 | 133 | 71.1%     |

Figure. Classifier performance



## Understanding the numbers

**Odds ratio (OR).** How the odds of the outcome multiply per one-unit increase in the predictor ( $>1$  raises the odds,  $<1$  lowers them). Here,  $OR = 1.04$ , so a one-unit increase in the predictor multiplies the odds by this factor.

**Log-odds (B).** The raw coefficient on the log-odds scale that the odds ratio is exponentiated from. Here,  $B = 0.04$  on the log-odds scale.

**Omnibus  $\chi^2$ .** Tests the model as a whole against a model with no predictors. Here, the omnibus  $\chi^2 = 31.41$  ( $p < .001$ ).

**Nagelkerke  $R^2$ .** A pseudo- $R^2$  approximating the share of variance explained (0 to 1, higher is more). Here, Nagelkerke  $R^2 = 0.17$ .

**Log.Lik..** The log-likelihood: how probable the observed outcomes are under the fitted model; it underlies AIC and BIC. Here,  $\text{Log.Lik.} = -141.46$ .

**AIC.** A fit measure that penalizes model complexity, useful for comparing models fit to the same data (lower is better). Here,  $AIC = 292.91$ .

**BIC.** Like AIC but penalizes complexity more heavily as sample size grows (lower is better). Here,  $BIC = 310.31$ .

**N.** The number of observations used to fit the model. Here,  $N = 240$ .

**Predicted \ Observed (first category).** The diagonal (correct) cell for the first outcome category: how many cases the model predicted into it that were actually observed there. Here, 33 cases were predicted and observed in the first category.

**Predicted \ Observed (second category).** The diagonal (correct) cell for the second outcome category: how many cases the model predicted into it that were actually observed there. Here, 133 cases were predicted and observed in the second category.

**% correct.** The overall share of cases the model classified correctly - model fit and the ROC/AUC show how well it classifies overall. Here, the model classified 69.2% of cases correctly.

**Accuracy.** The overall share of cases classified correctly (same number as % correct, reported as a named scalar here). Here, accuracy = 69.2%.

**Sensitivity.** The share of ACTUAL events the model correctly predicted as events (the true positive rate). Here, sensitivity = 86.9%.

**Specificity.** The share of ACTUAL non-events the model correctly predicted as non-events (the true negative rate). Here, specificity = 37.9%.

**AUC [95% CI].** The area under the ROC curve, with its confidence interval (pROC's DeLong method) - 0.5 is chance, 1.0 is perfect discrimination. Here, AUC = .71 [.64, .78].

### How to read this test

Each predictor's odds ratio tells how the odds of the outcome change per unit (>1 increases, <1 decreases); a term carries weight when its OR confidence interval excludes 1, and the inline omnibus  $\chi^2$  (with its p) tests the model as a whole. Model fit and the ROC/AUC show how well it classifies overall, alongside accuracy (overall % correct), sensitivity (% of actual events correctly predicted), and specificity (% of actual non-events correctly predicted). Your significance threshold ( $\alpha$ ) is 0.05.

**APA template:** Predictor pretest\_score was associated with the outcome, OR=1.04, 95% CI [1.01, 1.08], p = .025 (AUC=.71).

**Statistical basis:** Logistic regression coefficients (log-odds / odds ratios) (R Core Team (2026). R: A Language and Environment for Statistical Computing. R Foundation for Statistical Computing, Vienna.); Nagelkerke pseudo- $R^2$  (Lüdtke D, Ben-Shachar MS, Patil I, Waggoner P, Makowski D (2021). "performance: An R Package for Assessment, Comparison and Testing of Statistical Models." Journal of Open Source Software, 6(60), 3139.); ROC curve / AUC (Robin X, Turck N, Hainard A, et al. (2011). "pROC: an open-source package for R and S+ to analyze and compare ROC curves." BMC Bioinformatics, 12, 77.). Full references in the exported CITATIONS.txt.

After export · optional feedback - an anonymous, optional satisfaction survey - opens a short Google Form in a new tab

[How did it go? - Share feedback →](#)